Proving segment relationships

1

a Always

b Never

c Always

d Sometimes

2 No, C does not need to be on $\overrightarrow{AB}$.

3 $x = 2$

4 $x = 6$

5 Example answer using a two-column proof:

To prove: $PQ = PS - QS$

	Statements	Reasons
1.	Points P, Q, R, and S are collinear	Given
2.	$PS = PQ + QS$	Segment addition postulate
3.	$PS - QS = PQ$	Subtraction property of equality
4.	$PQ = PS - QS$	Symmetric property of equality

6 Example answer using a two-column proof:

To prove: $\overline{AC} \cong \overline{XZ}$

	Statements	Reasons
1.	$\overline{AB} \cong \overline{XY}$ and $\overline{BC} \cong \overline{YZ}$	Given
2.	$AB = XY, BC = YZ$	Definition of congruent segments
3.	$AB + BC = XY + YZ$	Addition property of equality
4.	$AB + BC = AC, XY + YZ = XZ$	Segment addition postulate
5.	$AC = XZ$	Substitution property of equality
6.	$\overline{AC} \cong \overline{XZ}$	Definition of congruent segments

7 Example answer using a two-column proof:

To prove: $2AB = AC$

	Statements	Reasons
1.	B is the midpoint of $\overline{AC}$	Given
2.	$\overline{AB}$ is congruent to $\overline{BC}$	Definition of midpoint
3.	$AB = BC$	Definition of congruent segments
4.	$AB + BC = AC$	Segment addition postulate
5.	$AB + AB = AC$	Substitution property of equality
6.	$2AB = AC$	Combine like terms

8 Example answer using a two-column proof:

To prove: $\overline{RS} \cong \overline{TU}$

	Statements	Reasons
1.	$\overline{RT} \cong \overline{SU}$	Given
2.	$RT = SU$	Definition of congruent segments
3.	$RT + ST = RT, ST + TU = SU$	Segment addition postulate
4.	$RS + ST = ST + TU$	Substitution property of equality
5.	$RS = TU$	Subtraction property of equality
6.	$\overline{RS} \cong \overline{TU}$	Definition of congruent segments